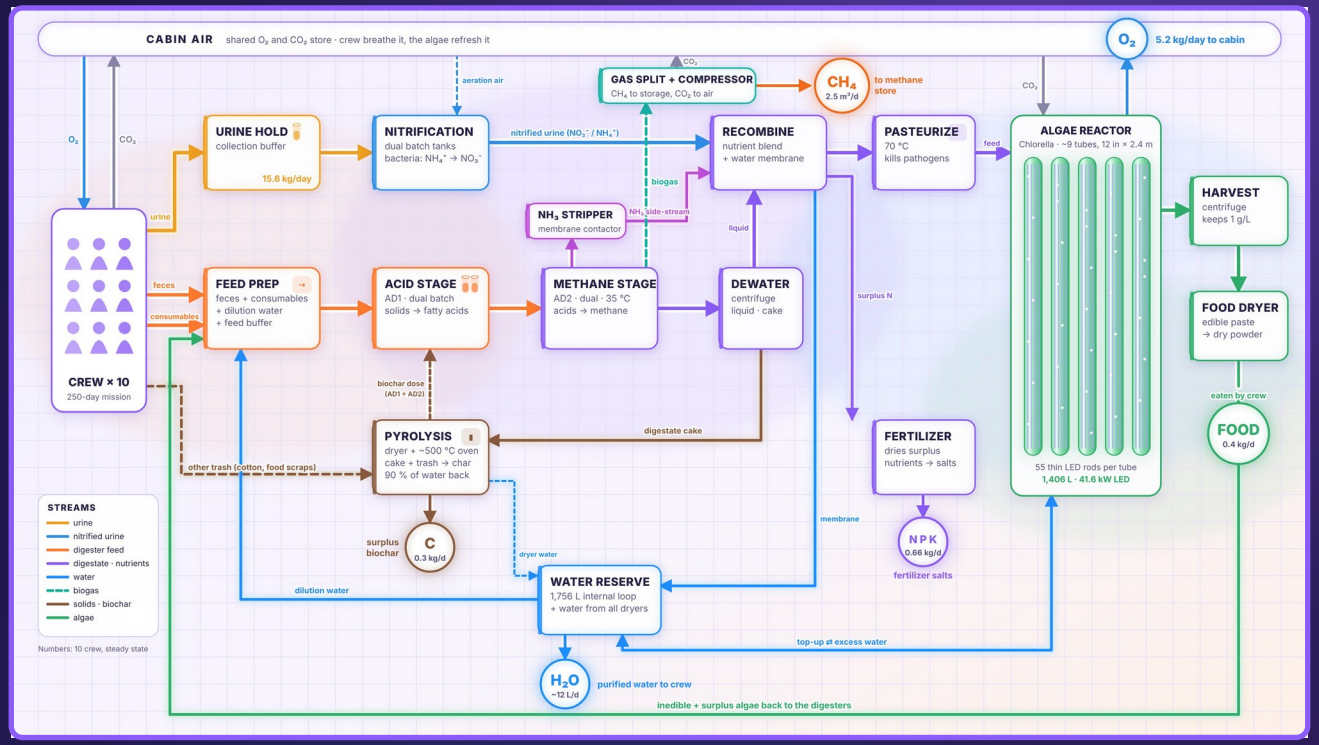
BIOREGENERATIVE LIFE SUPPORT

Closed-Loop Waste Recovery System

Using the microbes and algae Earth already relies on to turn crew waste into oxygen, food, water and fuel

10 CREW | 250 DAYS | 60% OF O_2



System overview · November 2026

Today, spaceflight stores its waste and then burns it



1 Bag it

Food packaging, clothing and wipes are sorted into wet and dry trash bags; solid human waste is sealed in containers



2 Store it

Bags wait aboard for months, taking up volume and risking odor and microbial growth



3 Load it

When a cargo ship is ready to leave, the crew pack it with trash



4 Burn it

The ship de-orbits and burns up in Earth's atmosphere, waste and all



Water is the exception

The ISS now recovers 98% of the water crews bring, including urine and sweat. Everything else is thrown away.



About 2,500 kg of trash a year

from four ISS astronauts. NASA's planned trash compactor squeezes it into tiles and recovers water; it doesn't recover anything else.

Sources: NASA water recovery milestone (2023); Nanoracks Bishop airlock trash test (2022); NASA Trash Compaction and Processing System.

For 10 crew on a 250-day trip, that's over 8 tonnes

3.9 t

of urine

1.56 kg per person per day

0.31 t

of feces

0.12 kg per person per day

4.0 t

of other trash

1.6 kg per person per day: packaging, wipes, clothing

No “away” beyond Earth orbit

No departing cargo ship to burn it in; it rides along for the whole trip.

Mass and volume

Every kilogram is launched, then stored in pressurized space someone could live in.

Odor and microbes

Wet waste keeps growing bacteria and must be sealed, dried or sterilized.

Thrown-away resources

The carbon, nitrogen, phosphorus and water in it all had to come from Earth.

Waste rates: NASA Life Support Baseline Values and Assumptions (BVAD). A 4-person, 180-day trip would make ~3,000 compacted trash bundles (AIAA Aerospace America).

On Earth, microbes do this work at room temperature

Every waste stream on Earth is someone else's food. Bacteria, fungi and algae break down organic matter, release its nutrients and rebuild them into new life, all at ordinary temperature and pressure. Industry often needs heat and pressure to do the same chemistry.

JOB	HOW BIOLOGY DOES IT	HOW MACHINES DO IT
Fixing nitrogen from air	Nitrogenase enzymes in bacteria: room temperature, 1 atm	Haber–Bosch: 400–500 °C, ~200 bar, 1–2% of world energy
Breaking down organic waste	Digester microbes at 35 °C, making methane fuel as they go	Incineration or pyrolysis at hundreds of °C
Turning CO_2 into oxygen	Algae and plants run on light	Water electrolysis, with CO_2 handled separately
Recycling nitrogen in urine	Nitrifying bacteria turn ammonia into nitrate plants can use	Distill the water out and throw the brine away

Haber–Bosch conditions and energy share: Frontiers in Chemistry (2021); Utah State University dissertation on nitrogenase. Digestion at 35 °C: this system's design.

Off Earth, our waste is a rare and useful material

ORDER → DISORDER

Food

highest stored energy

Crew waste

still energy-rich, nutrients concentrated

Methane + CO_2 + nutrients

microbes harvest energy, free the nutrients

CO_2, water, minerals

fully broken down: highest entropy

Waste is not high entropy, and that's the point

It still holds chemical energy and concentrated carbon, nitrogen and phosphorus. Life on Earth runs by stepping matter down this ladder, capturing energy at each step. Burning or storing waste skips every step.

Earth is the only place that makes it

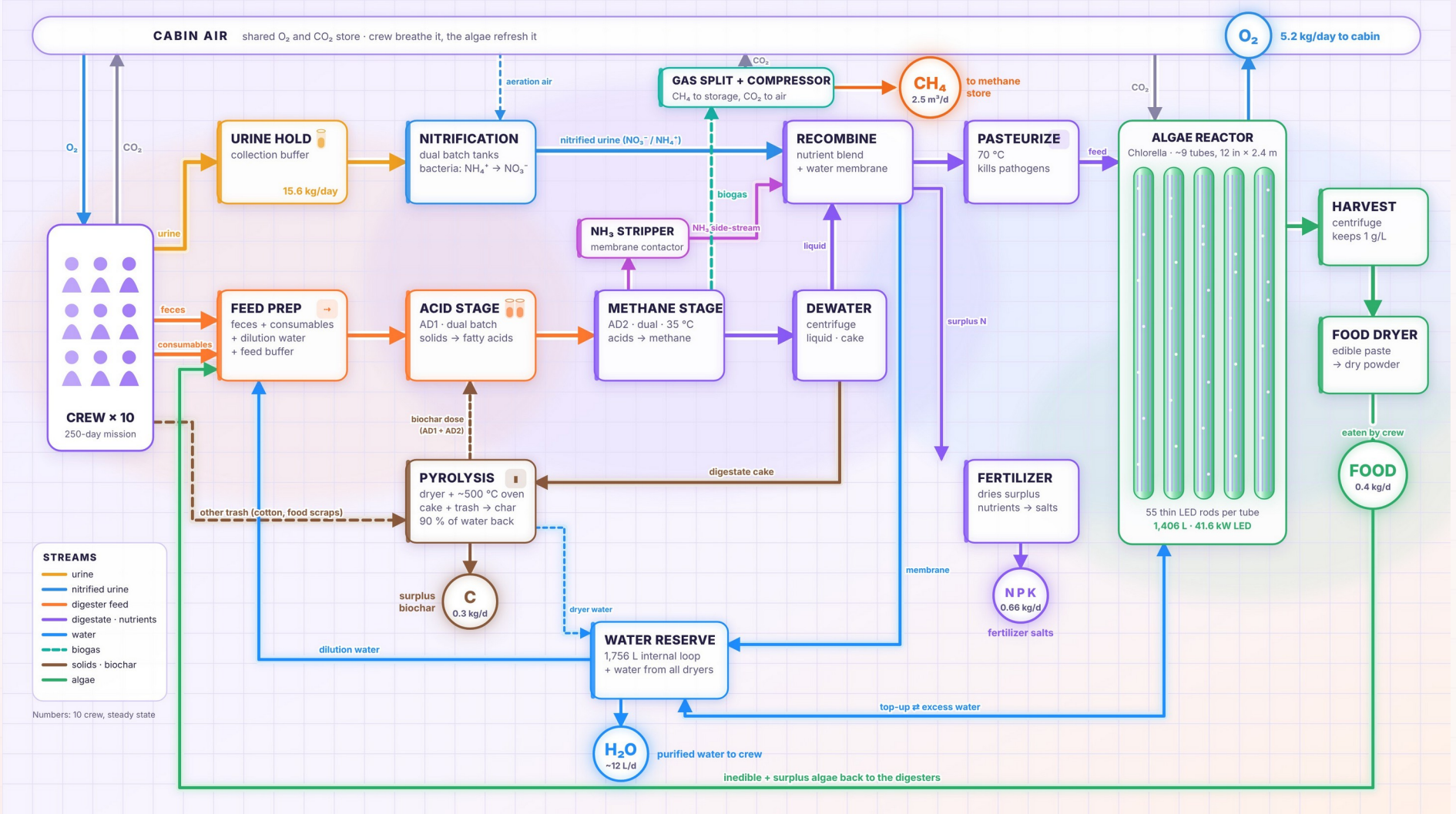
Organic carbon and fixed nitrogen are made by living things. Mars soil holds only traces of nitrate (up to ~0.1%) and very little organic carbon. In space, every gram had to be launched.

So biology can reuse it

The same microbes that close Earth's loops can turn the crew's waste back into air, food, water and fuel, instead of a burden to carry.

Mars nitrate: NASA/JPL, Curiosity SAM results (2015), up to 1,100 ppm. Life only existing on Earth: no life has been found anywhere else so far.

The system: four biological loops, one plant



Four stages, three of them alive



Nitrifying bacteria

EATS

Urine

Turn urine's ammonia into nitrate the algae can use. They take a little cabin O_2 to do it.



Methane-making microbes

EATS

Feces, paper, wipes, PHA, spare algae

Two-stage digestion at 35 °C turns solids into fatty acids, then methane fuel: 2.5 m^3 a day.



Pyrolysis (not biology)

EATS

Digestate cake, cotton, food scraps

Dried and heated into biochar. The char goes back into the digesters and speeds them up.



Chlorella algae

EATS

Nutrients + CO_2 + light

Make the crew's oxygen and edible protein, and pull water back through the loop.

Numbers from the simulation of the reference design (10 crew, steady state). Algae performance set to ISS-type Chlorella levels.

Closed-loop life support has been tested for 50 years

1972–84

BIOS-3

Krasnoyarsk. Up to 3 people sealed in for as long as 180 days; Chlorella under lamps recycled their air.

1991–93

Biosphere 2

Arizona. 8 people for 2 years. O_2 fell from 20.9% to 14.5%: soil microbes ate it. Lesson: balance every loop.

1995–now

MELiSSA pilot plant

Barcelona (ESA). Linked bioreactors, sized to make one person's oxygen, with rats as the test crew.

2017–18

Lunar Palace 1

Beijing. 370 days, 98% closure; 99.7% of urine and 67% of solid waste recovered.

2019–20

On the ISS

Chlorella reactor flown (PBR@LSR); ground twin ran 188 days at up to 1.3 g/L/day. Microbes mined basalt in orbit (BioRock).

Sources: Salisbury et al. 1997; Severinghaus et al. 1994; Garcia-Gragera et al. 2021; Fu et al. (Lunar Palace 365); Helisch et al. 2020; Cockell et al. 2020. Details in notes.

On Earth, turning waste into methane is routine

~9,900

biogas plants in Germany

Mostly on farms; Germany makes about half the EU's biogas

~40 million

household digesters in rural China

Cooking gas from household and animal waste (2010)

0.47 m³

methane per kg of solids

Measured at a full-scale plant digesting human feces



What's different in space

Earth's digesters are built for big, steady feeds and can vent or flare problems. A spacecraft digester is small, sealed, has to run for years without failing, and feeds an algae reactor and a crew instead of a gas grid. The microbes are the same; the engineering around them is the new part.

Sources: Fachverband Biogas via Bayern Innovativ (9,876 plants, 2022); IEA via Inkwood Research; Global Methane Initiative (38 million, 2010); Zhang et al. 2017, Int J Agric & Biol Eng.

What 10 crew get back over 250 days

oxygen

1,270 kg

about 60% of what the crew breathe

methane

591 m³

424 kg, about 1,000 L as liquid

purified water

2,560 L

surplus sent to the crew

algae grown

873 kg

100 kg eaten; the rest recycled

CO_2 captured

1,610 kg

taken back out of the cabin air

fertilizer salts

134 kg

carrying 39 kg of nitrogen

biochar

95 kg

mostly reused in the digesters; 38 kg spare

waste used up

5.1 t

urine, feces, paper, wipes, PHA (dry)

Simulation of the reference design, counted from a cold start (the first weeks are spent building up the algae and digesters). Steady-state rates are about 10% higher.

The same reactors can make much more



Bioplastic from methane

Methane-eating bacteria store PHB plastic, up to ~55% of their dry weight. It can be 3D-printed into parts, and digested back to methane when worn out.



Mining with microbes

On the ISS, bacteria leached rare earth elements out of basalt, the rock that covers the Moon and Mars, in microgravity and Mars gravity.



Fertilizer for crops

The dried nutrient salts and biochar are what a greenhouse needs, so the loop can grow into plants and fresh food.



Fuel feedstock

Stored methane can heat, power or top up propellant, and is the starting material for the bioplastic route.

Sources: PHB from biogas methane (Microbiology Society case study); methane-to-PHB for space (DoD Purview 2017; Univ. of Florida preprint 2026); BioRock (Cockell et al., Nature Communications 2020).

Where biology wins, and what it costs

What biology gives you

- Works at room temperature and pressure: no furnaces or high-pressure reactors to break down waste
- Does several jobs at once: one algae culture makes oxygen and food, removes CO_2 and recycles nitrogen
- Turns waste into fuel: about 2.5 m^3 of methane a day
- Saves launch mass: food, oxygen and water no longer all come from Earth

What it costs

- Electricity: with LED light, algae oxygen takes about 190 kWh per kg. Splitting water needs about 5 kWh per kg. Biology doesn't save energy; it saves mass.
- Living systems drift: Biosphere 2 lost oxygen to soil microbes. Every loop needs monitoring and control.
- Slow start-up: the digesters and algae take weeks to build up.
- On Mars, sunlight could supply part of the light and cut the power bill.

LED figure: 41.6 kW of LEDs for 5.2 kg O_2/day in the reference design. Electrolysis: thermodynamic minimum for splitting water, about 4–5 kWh per kg of O_2.

At full scale: a city of 10,000 on Mars

The 10-crew design multiplied by 1,000 (no economies of scale assumed)

5.2 t

of oxygen a day

60% of the city's breathing

665 t

of methane a year

≈ 2.8 Starship methane loads

11,400 L

of clean water a day

back from the waste loop

400 kg

of algae food a day

plus 3.1 t recycled to the digesters

240 t

of fertilizer salts a year

for greenhouse crops

20 t

of waste used up a day

urine, feces, paper, wipes, PHA

Hardware: about 1,400 m^3 of algae culture in ~9,300 tubes, 8,000 m^3 of tanks and 42 MW of LEDs, part of which Martian sunlight could replace.

Starship carries about 240 t of methane per full load (arXiv:2205.08567). Linear scale-up of the simulated 10-crew design at steady state.